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VEHICLE ARRESTING APPARATUS

Abstract

A vehicle arresting apparatus for launching a puncturing device onto a driving surface to puncture a motor vehicle's tires includes a projectile and a projectile launcher configured to launch the projectile. A deflation member is coupled to the projectile. The deflation member comprises a plurality of puncture members and a plurality of flexible links which are coupled together in alternating fashion. Each puncture member is configured to puncture a tire, and each flexible link couples a first end of one of the puncture members to a second end of another of the puncture members.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC
OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

[0004] Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT
INVENTOR

[0005] Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

[0006] The disclosure relates to vehicle arresting apparatuses and more particularly pertains to a new vehicle arresting apparatus for launching a puncturing device onto a driving surface to puncture a motor vehicle's tires.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

[0007] The prior art describes myriad vehicle arresting devices, including devices which deploy a piercing member across a road surface from one side in order to deflate a motor vehicle's tires. U.S. Pat. No. 10,030,343, for example, discloses a launching device which casts a flexible net with attached spikes across a road surface. The prior art fails, however, to disclose an apparatus which launches a projectile from a projectile launcher that draws a deflation member comprised of alternating rigid spiked members and flexible connectors across a driving surface.

BRIEF SUMMARY OF THE INVENTION

[0008] An embodiment of the disclosure meets the needs presented above by generally comprising a projectile and a projectile launcher configured to launch the projectile. A deflation member is coupled to the projectile. The deflation member comprises a plurality of puncture members and a plurality of flexible links which are coupled together in alternating fashion. Each puncture member is configured to puncture a tire, and each flexible link couples a first end of one of the puncture members to a second end of another of the puncture members.

[0009] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

[0010] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

[0011] The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0012] FIG. 1 is a side view of a vehicle arresting apparatus according to an embodiment of the disclosure.

[0013] FIG. 2 is a perspective detail view of a cartridge and a cartridge mount of an embodiment of the disclosure.

[0014] FIG. 3 is a detail view of a deflation member of an embodiment of the disclosure in a stored

configuration.

[0015] FIG. **4** is a detail view of a puncture member of a deflation member of an embodiment of the disclosure.

[0016] FIG. **5** is a detail view of a flexible link and a puncture member of a deflation member of an embodiment of the disclosure.

[0017] FIG. **6** is an in-use view of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0018] With reference now to the drawings, and in particular to FIGS. **1** through **6** thereof, a new vehicle arresting apparatus embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

[0019] As best illustrated in FIGS. **1** through **6**, the vehicle arresting apparatus **10** generally comprises a projectile **12** and a projectile launcher **14** configured to launch the projectile **12**. The projectile launcher **14** comprises a body **16** and a barrel **18**. The barrel **18** is coupled to and extends forwardly from the body **16** to a front end **20** which is open to facilitate release of the projectile **12** through the front end **20**. The projectile **12** is receivable into the barrel **18** and launchable out of the barrel **18** upon actuation of a firing mechanism of the projectile launcher **14**. The firing mechanism of the projectile launcher **14** may use explosive charges, compressed gas, linear induction motors, or any other suitable propellant for launching the projectile **12** from the projectile launcher **14**. In some cases, explosive charges or other energy stores may be coupled to the projectile **12** prior to launching the projectile **12**.

[0020] A deflation member **22** is coupled to the projectile **12** via a tether attached to a leading end **24** of the deflation member **22**. The deflation member **22** comprises a plurality of puncture members **28** and a plurality of flexible links **42** which are coupled together in alternating fashion. It is understood that the term “flexible link” is used in the present disclosure to refer to flexibility afforded the deflation member **22** which allows the deflation member to bend at the location of the flexible link. The flexible link may in fact comprise a rigid material such that the flexible link itself does not bend or otherwise deform.

[0021] The puncture members **28** are substantially rigid such that they maintain their shapes prior to use in puncturing a tire **72** of a motor vehicle **70**. Each flexible link **42** couples a first end **30** of one of the puncture members **28** to a second end **32** of another of the puncture members **28**. The puncture members **28** are elongated between the first and second ends **30**, **32** such that the deflation member **22** may be placed into a linear configuration **50** in which the puncture members **28** align with each other along a central axis of the deflation member **22**.

[0022] Each puncture member **28** comprises a capsule **34** and a plurality of quills **36**. The quills **36** are housed by the capsule **34** and are attached to each other by a rod **38** or flexible line that extends between the first and second ends **30**, **32** of the puncture member **28**. The capsule **34** comprises a material with a strength sufficient to rigidly maintain its shape until a weight force of the motor vehicle **70** is applied, at which point the capsule **34** splits and exposes one or more quills **36** from within the capsule **34**. The material of the capsules **34** may be a rigid plastic or a ceramic which fractures under the weight force, a flexible plastic or the like, which collapses and allows one or more quills **36** to pierce the capsule **34**, or any other suitable material. Each quill **36** has pointed ends configured to pierce one of the tires **72** of the motor vehicle **70** and is hollow. When any of the quills **36** becomes embedded in the tire **72**, the quill **36** facilitates passage of air from the tire **72** to a surrounding atmosphere.

[0023] Each flexible link **42** comprises a clip **44** with a gate **46** biased to form a closed loop. Each puncture member **28** comprises at least one attachment loop **40** which is interlocked with one of the flexible links **42**. The clip **44** may be a carabiner, or the flexible link **42** may comprise other suitable connection means such as a quick link chain link, a cable, or the like. The clip **44** may be opened to separate the adjacent puncture members **28** from each other for removal, replacement, or repair. Other embodiments also may have flexible links **42** which do not readily detach from the

adjacent puncture members **28**. Most of the puncture members **28** will have two attachment loops **40** for attaching to flexible links **42** positioned on either of the first and second ends **30**, **32** of the puncture member **28**. But the puncture members **28** at the leading end **24** and a trailing end **26** of the deflation member **22** may only have one attachment loop **40** for attaching to an adjacent one of the flexible links **42**.

[0024] A cartridge **54** is mounted to the projectile launcher **14** which contains the deflation member **22** in a stored configuration **48**. The deflation member **22** follows a serpentine path when in the stored configuration **48** wherein the puncture members **28** are arranged along a path substantially parallel to the barrel **18**. The puncture members **28** also lie parallel to each other and perpendicular to the path when in the stored configuration **48**. The cartridge **54** has an open forward end **56** such that the deflation member **22** may be removed forwardly from the cartridge **54**. The tether **52** extends through the open forward end **56** of the cartridge **54** when the deflation member **22** is in the stored configuration **48** so the projectile **12** may be positioned within the barrel **18** of the projectile launcher **14** when the cartridge **54** stores the deflation member **22**.

[0025] A cartridge mount **58** removably couples the cartridge **54** to the barrel **18** of the projectile launcher **14**. The cartridge mount **58** comprises a first mating member **60** coupled to the projectile launcher **14** and a second mating member **62** coupled to the cartridge **54**. The first mating member **60** comprises a rail **64**, and the second mating member **62** comprises a slider **66**. The rail **64** is oriented parallel to the barrel **18** and has a T-shaped cross section. The slider **66** defines a cavity **68** complementary in shape to the rail **64**. Threaded fasteners, locking pins, or the like may be used to secure the slider **66** to the rail **64**. In other embodiments, the cartridge mount **58** may comprise another suitable fastener including a bracket and threaded fasteners, a clamp, a snap connection, or the like.

[0026] In use, the cartridge **54** is coupled to the projectile launcher **14** via the cartridge mount **58** while containing the deflation member **22** in the stored configuration **48**. The projectile **12** is placed in the barrel **18** of the projectile launcher **14**. To deploy the deflation member **22** across a driving surface **74**, the firing mechanism of the projectile launcher **14** is actuated to launch the projectile **12**, which draws the deflation member **22** via the leading end **24** away from the projectile launcher **14**. As the deflation member **22** is drawn out of the cartridge **54**, each puncture member **28** is pulled into alignment with the other puncture members **28** and generally parallel to the direction of travel of the projectile **12**. The deflation member **22** is thereby positioned in the linear configuration **50** by nature of the operation of the projectile launcher **14** and falls to rest on the driving surface **74**.

[0027] When the motor vehicle **70** drives over one or more of the puncture members **28** of the deflation member **22**, the capsules **34** bearing the weight of the motor vehicle **70** split, and the quills **36** therein puncture the tires **72** of the motor vehicle **70**. The quills **36** become embedded in the tires **72**, which allows air to be forced out of the tires **72** via the pressure differential between the internal chambers of the tires **72** and the surrounding atmosphere. By deflating the tires **72**, the motor vehicle **70** loses traction with the driving surface **74** and is thus immobilized or decelerated by the vehicle arresting apparatus **10**.

[0028] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

[0029] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the

scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. A vehicle arresting apparatus comprising: a projectile; a projectile launcher configured to launch the projectile; and a deflation member coupled to the projectile, the deflation member comprising a plurality of puncture members and a plurality of flexible links, the puncture members and the flexible links being coupled together in alternating fashion, each flexible link coupling a first end of one of the puncture members to a second end of another of the puncture members, each puncture member being configured to puncture a tire.
2. The apparatus of claim 1, wherein the projectile launcher comprises a body and a barrel, the barrel being coupled to and extending forwardly from the body, the barrel being open at a front end of the barrel, the projectile being receivable into the barrel and launchable out of the barrel upon actuation of the projectile launcher.
3. The apparatus of claim 2, further comprising a cartridge mounted to the projectile launcher, the cartridge containing the deflation member in a stored configuration, the deflation member being removable forwardly from the cartridge upon launching the projectile via the projectile launcher.
4. The apparatus of claim 3, the deflation member following a serpentine path when in the stored configuration wherein the puncture members lie along a path substantially parallel to the barrel.
5. The apparatus of claim 3, further comprising a tether coupling the deflation member to the projectile, the tether being attached to a leading end of the deflation member, the tether extending through an open forward end of the cartridge when the deflation member is in the stored configuration.
6. The apparatus of claim 3, further comprising a cartridge mount removably coupling the cartridge to the barrel of the projectile launcher, the cartridge mount comprising a first mating member coupled to the projectile launcher and a second mating member coupled to the cartridge.
7. The apparatus of claim 6, wherein the first mating member comprises a rail, the second mating member comprising a slider, the rail being oriented parallel to the barrel.
8. The apparatus of claim 7, wherein the rail has a T-shaped cross section, the slider defining a cavity complementary in shape to the rail.
9. The apparatus of claim 1, wherein each puncture member is substantially rigid.
10. The apparatus of claim 1, wherein each puncture member comprises a capsule and a plurality of quills, the plurality of quills being housed by the capsule, the capsule being configured to split under a weight of a motor vehicle, each quill having pointed ends configured to pierce the tire.
11. The apparatus of claim 10, wherein each quill is hollow wherein each quill facilitates a passage of air from the tire to an atmosphere when the quill is embedded in the tire.
12. The apparatus of claim 1, wherein each flexible link comprises a clip with a gate biased to form a closed loop.
13. The apparatus of claim 1, wherein each puncture member comprises an attachment loop which is interlocked with one of the flexible links.
14. The apparatus of claim 1, further comprising a tether coupling a leading end of the deflation member to the projectile.
15. The apparatus of claim 1, further comprising a cartridge mounted to the projectile launcher, the cartridge containing the deflation member in a stored configuration, the cartridge having an open forward end such that the deflation member may be removed through the open forward end.
16. The apparatus of claim 15, wherein the deflation member follows a serpentine path when in the

stored configuration.

17. The apparatus of claim 16, wherein the puncture members are elongated and lie parallel to each other when the deflation member is in the stored configuration.

18. The apparatus of claim 15, further comprising a cartridge mount removably coupling the cartridge to the projectile launcher, the cartridge mount comprising a first mating member coupled to the projectile launcher and a second mating member coupled to the cartridge.

19. The apparatus of claim 18, wherein the first mating member comprises a rail, the second mating member comprising a slider, the rail having a T-shaped cross section, the slider defining a cavity complementary in shape to the rail.

20. A vehicle arresting apparatus comprising: a projectile; a projectile launcher configured to launch the projectile, the projectile launcher comprising a body and a barrel, the barrel being coupled to and extending forwardly from the body, the barrel being open at a front end of the barrel, the projectile being receivable into the barrel and launchable out of the barrel upon actuation of the projectile launcher; a deflation member coupled to the projectile, the deflation member comprising a plurality of puncture members and a plurality of flexible links, the puncture members and the flexible links being coupled together in alternating fashion, each flexible link coupling a first end of one of the puncture members to a second end of another of the puncture members, each puncture member being substantially rigid, each puncture member comprising a capsule and a plurality of quills, the plurality of quills being housed by the capsule, the capsule being configured to split under a weight of a motor vehicle, each quill having pointed ends configured to pierce a tire, each quill being hollow wherein each quill facilitates a passage of air from the tire to an atmosphere when the quill is embedded in the tire, each flexible link comprising a clip with a gate biased to form a closed loop, each puncture member comprising an attachment loop which is interlocked with one of the flexible links; a tether coupling a leading end of the deflation member to the projectile; a cartridge mounted to the projectile launcher, the cartridge containing the deflation member in a stored configuration, the deflation member following a serpentine path when in the stored configuration wherein the puncture members lie along a path substantially parallel to the barrel, the cartridge having an open forward end such that the deflation member may be removed through the open forward end, the tether extending through the open forward end of the cartridge when the deflation member is in the stored configuration, the puncture members being elongated and lying parallel to each other when the deflation member is in the stored configuration; and a cartridge mount removably coupling the cartridge to the barrel of the projectile launcher, the cartridge mount comprising a first mating member coupled to the projectile launcher and a second mating member coupled to the cartridge, the first mating member comprising a rail, the second mating member comprising a slider, the rail being oriented parallel to the barrel, the rail having a T-shaped cross section, the slider defining a cavity complementary in shape to the rail.
